As a pathologist, I studied disease through a microscope. As an epidemiologist, I studied patterns across communities.

Then it became personal.

After I lost my job, my family lost our health insurance. The next month, my husband was bitten by a tick and became ill with Lyme disease.

For immigrants like my husband, language can add another barrier: understanding tests, asking about bills, and explaining symptoms.

I built PathwayAI to help people prepare for care and make these hidden burdens visible.

I'm Yuwei Zhang. Let me show you.

PathwayAI follows the whole journey. Before a bite, people can prepare for time outdoors. After a bite, they can find removal steps and time-sensitive care guidance.

But knowing you need care is only the beginning.

Without insurance, where do you start? Here are community health centers, income-based fee questions, coverage resources, and hospital financial-assistance information. Care is not automatically free, but people have a practical next step.

Now imagine someone preparing for an appointment. This example is fictional.

They describe a tick encounter, symptoms two weeks later, and a blood test they report was positive. PathwayAI organizes a timeline they can review, correct, and download. Symptom onset and diagnosis timing stay separate. The clinician still needs the original test report. A reviewed record can help someone explain their experience across appointments, without starting from the beginning each time.

People can write in their preferred language. With permission, optional AI translates and organizes an English summary alongside the original. It supports preparation, not professional medical interpretation. The basic English organizer works without external AI.

The value is bringing these pieces into one usable workflow, rather than asking people to navigate every source separately.

Now look beyond the visit. This person paid two hundred twenty-five dollars for care and travel, missed three workdays, and received eight hours of unpaid help.

Money, work, and caregiving stay separate. Unknown answers stay unknown.

They can prepare a request about transport eligibility, cost, booking, and the return trip. Someone facing a bill can record the unpaid balance and prepare assistance questions.

Now we move from one household to the community.

Dutchess and Orange county evidence and published cost research provide context. The county cost illustration is a scenario, not measured spending.

The downloadable planning brief helps partners discuss a practical response, staffing, funding, and what to track: Did people reach care? Did assistance come through? Who still needs help?

Early treatment can prevent serious complications; reducing delays, disruption, and costs is the goal we will evaluate.

Our main planning users are county public-health and community teams. Patients and caregivers receive preparation tools directly. Dutchess and Orange are our evidence examples, not committed county partners.

Federal sources include CDC surveillance, Census population context, HRSA healthcare resources, and CDC PLACES. New York surveillance and published research add separate evidence. AI organizes language; arithmetic calculates costs.

Advocate discussions highlighted unpaid care and daily difficulties. External public-health feedback pushed us to make practical next steps clearer. We improved summaries, language assistance, bill preparation, and transport guidance.

Sixty-four automated tests passed. These verify software behavior. Formal usability studies, clinical translation accuracy, improved access, and savings remain to be evaluated.

The pilot web app is already available on Streamlit.

My proposed next step is a small pilot recruited through willing patient groups, health centers, and care navigators. Shared patient-report collection stays off until consent, secure storage, and follow-up are ready.

The proposed sustainability model is free patient access supported by grants and community partnerships to cover hosting and AI costs. That funding is not yet secured.

Next, we will test usefulness, verify resources, and evaluate support connections before expanding. Over time, repeated records could help us study longer-term burden.

We are seeking patients, caregivers, clinicians, and community partners to help.

PathwayAI: Full Economic Burden. Making the Hidden Patient Journey Visible.

Thank you.